

THE ASSOCIATION OF MATHEMATICS TEACHERS OF INDIA

NMTC at JUNIOR LEVEL - IX & X Standards

Saturday, 12 February, 2022

Time : 2 Hrs.

M. M. : 30

Note:

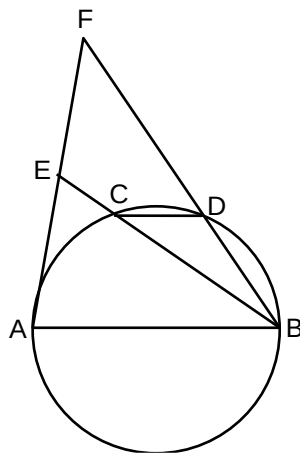
1. Fill in the response sheet with your Name, Class and the institution through which you appear in the specified places.
 2. Diagrams are only visual aids; they are NOT drawn to scale.
 3. You are free to do rough work on separate sheets.
 4. Duration of the test: **2 hours**.
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PART—A

Note

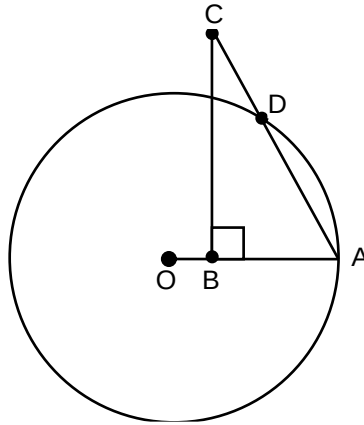
- Only one of the choices A, B, C, D is correct for each question. Shade the alphabet of your choice in the response sheet. If you have any doubt in the method of answering, seek the guidance of the supervisor.
 - For each correct response you get 1 mark. **For each incorrect response you lose $\frac{1}{2}$ mark.**
1. In an arithmetical progression the n^{th} term is t_n . If $t_1 + t_3 = t_4 - 2$, $t_6 + t_4 = t_{11} - 7$, then the value of t_{2021} is
(a) 10103 (b) 100103 (c) 1003 (d) 1000
 2. P, Q are the mid points of sides CA and CB respectively of a right triangle ABC, whose hypotenuse is AB. Then the numerical value $\frac{AQ^2 + BP^2}{AB^2}$ is
(a) $\frac{4}{5}$ (b) $\frac{3}{4}$ (c) $\frac{5}{4}$ (d) $\frac{1}{5}$
 3. If the zeroes of the polynomial $x^2 - px + q$ are in the ratio 2 : 3, the value of $\frac{p^2}{q}$ is
(a) $\frac{25}{6}$ (b) $\frac{24}{5}$ (c) $\frac{25}{3}$ (d) $\frac{25}{7}$
 4. If $\tan \theta - \cot \theta = m$ and $\cos \theta + \sin \theta = n$, then the value of $(m^2 + 4)(n^2 - 1)^2$ is
(a) 1 (b) 2 (c) 3 (d) 4
 5. If x_1, x_2 satisfy $\sqrt{\frac{x}{1-x}} + \sqrt{\frac{1-x}{x}} = \frac{13}{6}$, then $\frac{x_1}{x_2}$ is equal to
(a) $\frac{9}{4}$ (b) $\frac{4}{9}$ (c) $\frac{9}{4}$ or $\frac{4}{9}$ (d) 1

6. If $\frac{1}{b+c} + \frac{1}{c+a} = \frac{2}{a+b}$, then the value of $\frac{a^2+b^2}{c^2}$ is
 (a) 1 (b) 2 (c) 3 (d) 4
7. The sum of the squares of the roots of the equation $\log_{3-4x^2} (9-16x^4) = 2 + \frac{1}{\log_2(3-4x^2)}$ is
 (a) 1 (b) 2 (c) 3 (d) $\frac{1}{2}$
8. The number of natural numbers 'n' such that $(n^3 - 8n^2 + 20n - 13)$ be a prime is
 (a) 1 (b) 2 (c) 3 (d) 4
9. If x is real and $x^5 - x^3 + x = a$, then the minimum value of x^6 is
 (a) 2a (b) $2a - 1$ (c) $2a + 1$ (d) 3a
10. The maximum number of right angles which can occur among interior angles of a convex polygon of n sides, where $n > 4$ is
 (a) 2 (b) 3 (c) 4 (d) 5
11. In the adjoining diagram, CD is parallel to the diameter AB of the circle of radius 5 units. Line segment AEF is tangent to the circle. The value of AE. AF (in Square units) is



- (a) 25 (b) 50 (c) 100 (d) 200
12. If $\frac{xy}{x+y} = \frac{4}{3}$, $\frac{yz}{y+z} = \frac{12}{5}$ and $\frac{zx}{z+x} = \frac{3}{2}$, then
 (a) $x + y = z$ (b) $y + z = x$ (c) $z + x = y$ (d) $y + x = z$
13. If $f(x) = x^{15} - 2022x^{14} + 2022x^{13} - 2022x^{12} + \dots + 2022x$, then that value of $f(2021)$ is
 (a) $(2021)^7$ (b) $(2021)^6$ (c) 0 (d) 2021
14. The sum of the squares of all the roots of the equation $2^{x^2-1} - 3^{x^2} = 3^{x^2-1} - 2^{x^2+2}$ is
 (a) 6 (b) 5 (c) $2\sqrt{3}$ (d) $4\sqrt{3}$

15. In the adjoining figure, O is the centre of the circle. CB is Perpendicular to OA. CA cuts the circle in D. The value of $\frac{CA \cdot AD}{OA \cdot AB}$ is



- (a) 1 (b) 2 (c) 3 (d) $\frac{1}{2}$

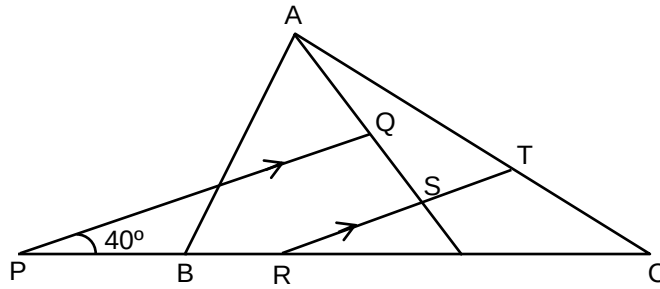
PART - B

Note :

- Write the correct answer in the space provided in the response sheet
- For each correct response you get 1 mark. For each incorrect response you lose $\frac{1}{4}$ marks.

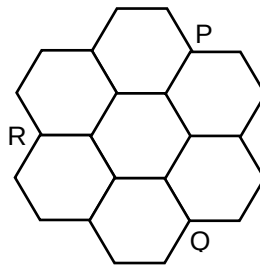
16. The roots of $(x + 1)^{\log(x+1)} = 100(x + 1)$ are x_1 and x_2 with $x_1 < x_2$. The value of $10x_1 + x_2$ is ____.
17. The numerical value of $\frac{1}{x^3} \{ (x + 5y - 3z)^3 + (x - 5y + 3z)^3 + 6x(x + 5y - 3z)(x - 5y + 3z) \}$ is ____.
18. P is a point outside an equilateral triangle ABC, but inside the region $\angle ABC$. If the distances from P to BC, CA and AB are h_1 , h_2 and h_3 respectively, and $h_1 - h_2 + h_3 = 6$ and the area of $\triangle ABC$ is $\lambda\sqrt{3}$ then the value of λ is ____.
19. Given G is the centroid of $\triangle ABC$. $GA = 2\sqrt{3}$, $GB = 2\sqrt{2}$ and $GC = 2$. If the area of $\triangle ABC$ is $\lambda\sqrt{2}$, then the value of λ is ____.
20. The number of pair of natural numbers (m, n) such that $(m^2 + n) - (m + n^2) = 10$ is ____.
21. The number of real values of n for which $\sqrt{120 - \sqrt{n}}$ is an integer is ____.
22. The number of cuboids with unequal integer side- length in arithmetic progression such that the volume is less than 25 cubic units is ____.
23. If a and b are real such that $a^2 + b^2 + 8a - 14b + 65 = 0$, then the numerical value of $(a^2 + b^2)$ is ____.
24. If a is real and $a^4 + a^3 + a^2 + a + 1 = 0$ then the numerical value of $a^{2022} + \frac{1}{a^{2022}}$ is ____.
25. If $a = 1$ and b, c, d > 0 are real such that $a^4 + b^4 + c^4 + d^4 = 4abcd$, then numerical value of $a^{2022} + b^{2022} + c^{2022} + d^{2022}$ is ____.
26. A circle is inscribed in a regular octagon. A and P are the area and perimeter of the octagon and R is the radius of the circle. Then the numerical value of $\frac{RP}{A}$ is ____.
27. In the adjoining figure ABC is an equilateral triangle.

PQ and RST are parallel and AS = ST.
 Also, $\angle CPQ = 40^\circ$ and $\angle BAT = x^\circ$.
 Then x is _____.



28. Then n^{th} term of a sequence is given by $t_n = \frac{(1 + (-1)^n) + 1}{3n + 1}$ Then the value of $\frac{1}{t_1} + \frac{1}{t_3} + \frac{1}{t_5}$ is _____.

29. Seven regular hexagons of equal sides of 1 unit are placed as in the figure. Then area of triangle PQR (in square units) is $a\sqrt{3}$.
 Then a is _____.



30. Two real numbers m, n ($m \neq 0$, $n \neq 0$) are such $mn = m - n$. Then numerical value of $\frac{m}{n} + \frac{n}{m} - mn$ is _____.